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SURGICAL RETRACTOR HAVING CLAMPING MECHANISM

Abstract

A clamp for a surgical retractor system. The clamp has a tapered conically shaped washer against and through which clamping force is applied to place the clamp in a clamped position. The clamping force is applied by a cam head connected to a central shaft. A locking pin connects the cam head to the shaft. Apertures are provided in the clamp for receiving rod ends of a surgical retractor. Such apertures are circular and are positioned to be closer to the top and bottom edges of the clamp portion of the clamp than to the end edge of the clamp portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. patent application Ser. No. 18/581,230, filed Feb. 19, 2024, which is a continuation of U.S. patent application Ser. No. 17/202,665, filed Mar. 16, 2021, which is a continuation of U.S. patent application Ser. No. 16/299,400, filed Mar. 12, 2019, which is a continuation of U.S. patent application Ser. No. 15/232,466, filed Aug. 9, 2016, which claims benefit to and priority to U.S. Provisional Patent Application No. 62/203,144, filed Aug. 10, 2015, the contents of each of which are hereby incorporated herein by reference in their entirety.

BACKGROUND OF THE INVENTION

[0002] The present invention relates to retractor systems for use during invasive surgical medical procedures.

[0003] During surgical medical procedures, a surgeon will typically make an incision in a patient to access the site of interest for the particular surgical procedure. To maintain clear access to the site of interest, a surgical retractor system is typically utilized. A surgical retractor system typically consists of a rail clamp, a frame connected to the rail clamp by a universal connecting joint mechanism, and retractor blades that are connected to the frame by additional universal connecting joint mechanisms. The rail clamp is commonly secured to an operating table and provides a fixed and sturdy support for the frame and the retractor blades. Each of the components in a typical surgical retractor system is conventionally made of stainless steel. The reason that stainless steel is generally used is that stainless steel is easily sterilized. As would be expected, before any use of the surgical retractor system can be made during a surgical procedure, the system must be thoroughly sterilized for the protection of the patient.

[0004] Surgical retractor systems have been made of other materials in the past, such as aluminum and titanium. The common characteristic of all of these materials is that they are highly durable and easily sterilized. That is, these materials are not porous and easily lend themselves to typical sterilization procedures used in modern day hospitals.

[0005] Accordingly, it is an object of the invention to provide a surgical retractor system that can be used confidently in surgical procedures.

[0006] It is a further object of the invention to provide a universal clamping mechanism for a surgical retractor system.

[0007] It is a further object of the invention to provide a clamping mechanism that is strong against wear caused by continual use of the clamping mechanism over time.

BRIEF SUMMARY OF THE INVENTION

[0008] These and other objects of the invention are achieved in a surgical retractor system which includes at least one clamp. The surgical retractor system may include a cam head having a flat latching surface, a tapered area washer configured to receive a central shaft and apply force to the at least one clamp, a locking pin configured to connect the central shaft to the cam head, and/or an offset applied to a passage used to clamp rods in the surgical retractor system.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

[0009] FIG. 1 is a perspective view of a typical surgical retractor system.

[0010] FIG. 2 is an elevated view of a rail clamp with a universal clamping mechanism of the present invention and a cross bar for insertion into the insulated universal clamping mechanism.

[0011] FIG. 3 is a perspective view of the universal clamping mechanism of the present invention with an insulating sleeve.

[0012] FIG. 4 is a side view of the universal clamping mechanism of FIG. 3 and in its open position.

[0013] FIG. 5 is a side view of the universal clamping mechanism of FIG. 3 and in its closed position.

[0014] FIG. 6 is a side view of the cam head of the clamping mechanism of FIG. 3.

[0015] FIGS. 7A, 7B and 7C are views of a washer and a clamp of the clamping mechanism of FIG. 3.

[0016] FIG. 8 is a side view of the washer of FIG. 7.

[0017] FIG. 9 is a side view of a washer of the clamping mechanism of FIG. 3.

[0018] FIG. 10 is a top view of the clamping mechanism of FIG. 3.

[0019] FIG. 11 is a cross sectional side view of the clamping mechanism of FIG. 9 along lines A-A.

[0020] FIG. 12 is a side view of a shaft of clamping mechanism of FIG. 3.

[0021] FIG. 13 is an exploded view of some components of the clamping mechanism of FIG. 3

[0022] FIG. 14 is an exploded view of some components of the clamping mechanism of FIG. 3

[0023] FIG. 15 is a side view of a second embodiment of a universal clamping mechanism.

[0024] FIG. 16 is a side view of a third embodiment of a universal clamping mechanism.

DETAILED DESCRIPTION OF THE INVENTION

[0025] In the following detailed description, spatially orienting terms are used such as “left,” “right,” “vertical,” “horizontal,” and the like. It is to be understood that these terms are used for convenience of description of the preferred embodiments by reference to the drawings. These terms do not necessarily describe the absolute location in space, such as left, right, upward, downward, etc., that any part must assume.

[0026] As shown in FIG. 1, a surgical retraction system 10 includes an adjustable rail clamp 12 which is securable to a conventional surgical table 16. A second adjustable rail clamp (not shown) may be secured to the opposite side of the surgical table for increased stability if desired or needed. A post 17 extends vertically from clamp 12 to provide support for a cross bar 18 which in turn provides support for a pair of extension arms 19, 20. Cross bar 18 is secured to post 17 by a multi-directional joint clamp 21. Extension arms 19, 20 are respectively secured to cross bar 18 by a pair of multi-directional joint clamps 22, 24. Additional joint clamps 26, 28 are disposed along extension arms 19, 20 for rigidly securing any number of retractor blades 34, 36 to extension arms 19, 20. As will suggest itself, extension arm 19 may be secured directly to post 17 by a joint clamp, thus in many circumstances eliminating the need for cross bar 18.

[0027] The purpose of this frame is to securely connect the retractor blades 34, 36 to the rail clamp 12 to eliminate unwanted relative movement between the retractor blades 34, 36 and the surgical table 16, this may be accomplished by connecting the retractor blades to the cross bar 18 directly or through extension arms 19, 20, or, as already noted, by connecting extension arms 19, 20 directly to the rail clamp.

[0028] Each retractor blade 34, 36 includes a blade portion 42 and a retractor arm 44. Blade portion 42 extends downwardly into the incision 46 made by the surgeon. Blade portion 42 is used to retract anatomy to make the incision 46 accessible to the surgeon.

[0029] FIG. 2 is an elevated view of the rail clamp 12 and cross bar 18. Of course, as noted above,

extension arm **19** or **20** may be connected directly to adjustable rail clamp **12**. The rail clamp includes a clamp **50** that may be secured to surgical table **16** (FIG. **1**). This may be a conventional clamp as presently used in the industry and provides for a secure attachment of the adjustable rail clamp **12** to the surgical table **16**. Multi-directional joint clamp **21** is shown at the upper most extremity of post **17**. This position of the multi-directional joint clamp **21** enables the user to locate cross bar **18** at a height sufficient for the surgical retractor system **10** (FIG. **1**) to be used during surgical procedures. FIG. **2** further shows joint clamps **22**, **24** in their position on cross bar **18**. Clamps **22** and **24** may be identical to each other or different depending on the intended use of each of those clamps. Additionally, these joint clamps **22**, **24** may each be the same as or different to the multi-directional joint clamp **21** on post **17**.

[0030] Referring to FIGS. **3-5** and **14**, multi-directional joint clamp **21** includes a first clamp **52** and a second clamp **54**. Joint clamp **21** includes a central shaft **56** as well as a cam lever **58** for bringing first clamp **52** and second clamp **54** into a clamping position or into an unclamping position. Joint clamp **21** may also include a washer **57** positioned between the cam lever **58** and the first clamp **52**.

[0031] The first clamp **52** may include a passage **70**, which is intended to accommodate, for example, post **17** of rail clamp **12** (FIG. **2**). Similarly, the second clamp **54** may include a passage **72**. Passage **72** is intended to accommodate, for example, cross bar **18** (FIG. **2**). This type of joint clamp is more fully described in U.S. Pat. Nos. 5,897,087 and 6,033,363, which are incorporated herein by reference.

[0032] The first cylindrical passage **70** of the first clamp **52** is defined by a broken cylindrical surface **71**. The cylindrical surface **71** is broken along two parallel legs **73**, **74** which run the axial length of the surface **71** to define a gap **75**. Legs **73**, **74** are movable with respect to one another in order to shorten the gap **75** and thus constrict the area circumscribed by cylindrical surface **71**. Each leg **73**, **74** defines a respective planar surface **76**, **77**. Each planar surface **76**, **77** is generally parallel to the longitudinal axis of cylindrical passage **70**. The application of a clamping forces directs forces against an upper surface **78a** and a lower surface **78b** of the first clamp **52** in a direction toward the gap **75** and parallel to the longitudinal axis of shaft **56** and serves to move planar surfaces **76**, **77** toward each other constricting the area within cylindrical passage **70**.

[0033] Similarly, the second cylindrical passage **72** of the second clamp **54** is defined by a broken cylindrical surface **80**. The cylindrical surface **80** is broken along two parallel legs **82**, **84** which run the axial length of the surface **80** to define a gap **86**. Legs **82**, **84** are movable with respect to one another in order to shorten the gap **86** and thus constrict the area circumscribed by cylindrical surface **80**. Each legs **82**, **84** defines a respective planar surface **90**, **92**. Each planar surface **90**, **92** is generally parallel to the longitudinal axis of cylindrical passage **80**. The application of a clamping force directs forces against an upper surface **94a** and a lower surface **94b** of the second clamp **54** in a direction toward the gap **86** and parallel to the longitudinal axis of shaft **56** and serves to move planar surfaces **90**, **92** toward each other constricting the area within cylindrical passage **72**.

[0034] The clamp **52**, **54** further include locking teeth. In particular, a first set of locking teeth **99b** is disposed on the lower surface **78b** of first clamp **52** and a second set of locking teeth **99a** is disposed on the upper surface **94a** of second clamp member **54**. The sets of locking teeth **99a**, **99b** engage each other when clamp members **52**, **54** are compressed together by cam lever **58**, thereby fixing the positions of clamps **52**, **54** relative to one another.

[0035] The cam lever **58** includes a cylindrical shaped handle **81** having a distal end **83** connected to a cam head **85**. Cam head **85** is pivotally mounted to central shaft **56** such that rotation of the cam lever **58** about a pivot axis **88** brings first clamp **52** and second clamp **54** into clamping positions and unclamping positions. The shaft **56** may be cylindrical in cross section and may extend through the cylindrical apertures **112**, **116** in clamp members **52**, **54**.

[0036] Referring now to FIG. **6**, an outer edge **100** of the cam head **85** is eccentric, i.e. multiply

curved, such that the distance D between the pivot axis **88** and the upper surface **91** of the washer **57** increases as the cam lever **58** is moved from its open position (e.g., FIG. **4**) towards its closed position (e.g., FIG. **5**). Hence, movement of cam lever **58** towards its closed position draws shaft **56** upwardly through cylindrical apertures **112**, **116**, compressing clamp members **52**, **54** between the outer edge **100** of the cam head **85** and a stop, such as the nut **113**, secured to the distal end of shaft **56**. Compression of the resilient clamp members **52**, **54** initially causes the sets locking teeth **99a**, **99b** to engage each other, thereby fixing the positions of the clamp members **52**, **54** relative to each other. Further compression of clamp members **52**, **54** constricts the area circumscribed by cylindrical passages **70**, **72** to secure clamps **52**, **54** to the arms passing through cylindrical passages **70**, **72**.

[0037] Rotation of cam lever **58** towards its open position moves shaft **56** downwardly through cylindrical apertures **112**, **116**. Due to the downward movement of shaft **56**, the nut **113** may be urged away from aperture **116**, thus reducing or removing a clamping force applied to resilient clamps **52**, **53**. At the open position, clamps **52**, **54** are loosely held on shaft **56**, and may be rotated relative to one another about shaft **56**. The nut **113** may prevent clamps **52**, **54** from being removed for the shaft **56**. The removed or reduced compression on clamps **52**, **54** may cause cylindrical passages **70**, **72** to expand their diameter, allowing clamps **52**, **54** to be moved relative to posts positioned in cylindrical passages **70**, **72**. The open position may also allow the various clamp components to be accessed for cleaning.

[0038] FIG. **4** shows the two clamps **52**, **54** in their unclamped positions. Cam lever **58** is rotated upwardly to an open position for releasing or untightening of clamps **52**, **54**. In this open position, a release surface **87** of cam head **85** is moved or rotated to a position above washer **57** providing little or no contact between cam head **85** and washer **57** and reducing the distance D between the pivot axis **88** and the upper surface **91** of the washer **57**. This removes the force on clamps **52**, **54** allowing them to expand and open their respective passages **70**, **72** to a full extent.

[0039] FIG. **5** shows the two clamps **52**, **54** in their clamped positions. Cam lever **58** is rotated downwardly to a closed position for tightening of clamp **52**, **54**. In this closed position, release surface **87** is rotated away from washer **57** and a flat planar surface **89** of cam head **85** engages washer **57**. To enter this closed position, a clamping force is applied to clamps **52**, **54** which constricts passages **70**, **72**.

[0040] Planar surface **89** serves to lock handle **81** into position. Because surface **89** is a flat planar surface and contacts the flat planar top surface **91** of the washer **57**, the reverse movement of cam lever **58** into an open position requires more force making it difficult for cam head **85** to unintentionally loosen and rotate away from its closed position of FIG. **5**.

[0041] FIG. **6** is a side view of cam head **85** illustrating the surfaces **87**, **89**. Surfaces **87**, **89** extend across the bottom of cam head **85**. Surfaces **87**, **89** are positioned on the cam head relative to axis **88** so as to engage the flat top surface **91** (FIG. **5**) of washer **57** when the cam lever **58** is pivoted.

[0042] Referring to FIGS. **7A-7C** and FIG. **8**, washer **57** is generally cylindrical in shape, and has a cone-like configured shape at its bottom. Washer **57** includes an outer cylindrical surface **101** and a truncated cone surface **103**. An aperture **105** extends fully through the washer terminating at the end of cone surface **103**. Aperture **105** has a diameter to receive central shaft **56** (FIG. **4**) which is cylindrical in shape.

[0043] Referring to FIGS. **4**, **10** and **11**, central shaft **56** extends into cam head **85** and is rotatably secured in place by a cam pivot pin **135**. This allows pivoting of cam lever **58** into its open position (FIG. **4**) and into its closed position (FIG. **5**).

[0044] Referring again to FIGS. **7A-7C**, clamp **52** has a conically shaped aperture **111** to receive and mate with the conical surface **103** of washer **57**. As shown in FIG. **7C**, conical aperture **111** meets with cylindrical aperture **112** which has a diameter to receive central shaft **56**. Similarly, clamp **54** has a conically shaped aperture **114** to receive and mate with the conical surface **119** of nut **113**. See, FIG. **14**. As shown in FIG. **14**, conical aperture **114** meets with cylindrical aperture

116 which has a diameter to receive central shaft **56**.

[0045] As shown in FIG. 4, a nut **113** is threadedly secured onto the distal end of shaft **56**. Threads **115** are shown in FIG. 12. Referring to FIG. 9, nut **113** is generally cylindrical in shape, having a cone-like configured shape at its top. Nut **113** includes a cylindrical surface **117** and a truncated cone surface **119**. An aperture **121** extends fully through nut **113** terminating at the end of conical surface **119**. Aperture **121** has a diameter to receive central shaft **56** and has threads **122** to mate with threads **115** on shaft **56** (FIG. 12).

[0046] The nut **113** may be welded to the shaft **56** after being threadedly secured to the distal end of the shaft **56**. Such welding may prevent further rotation of the nut **113** in relation to the shaft **56** about a longitudinal axis of the shaft **56**, but may permit rotation of the nut **113** with respect the clamps **52**, **54** about the longitudinal axis of the shaft **56**. Because nut **113** is rounded, the shaft **56** with welded nut **113** may be rotated 360°, even when nut **113** is partially drawn into aperture **116**. This may permit the surgeon to rotate handle **81** to a convenient position, even after clamps **52**, **54** have been compressed.

[0047] Prior to welding, the nut **113** may be rotated in one direction (e.g., clockwise) to tighten the nut **113** and increase a clamping force applied by the clamps **52**, **54** and may be rotated in the opposite direction (e.g. counter-clockwise) to loosen the nut **113** and decrease the clamping force applied by the clamps **52**, **54**. In this manner, the clamping force of the clamps **52**, **54** may be calibrated to a desired force. After obtaining the desired, calibrated clamping force, the nut **113** may be welded to ensure the nut **113** does not rotate further and the desired, clamping force is maintained.

[0048] Referring again to FIG. 3, cam head **85** is shaped to include two extending side members **123**, **125** forming an interposed channel **127**. Channel **127** is defined by two planar side surfaces **129**, **131** (FIG. 13) of the side members **123**, **125**. Each side member includes a cylindrical aperture **133** (FIG. 3) which are aligned along their axis. A cam pin **135** (FIG. 3) passes through cylindrical apertures **133**. Cam pin **135** is held fixed relative to cam head **85**.

[0049] As shown in FIG. 12, a top end **137** of the shaft **56** is generally planar in shape as best seen in FIG. 13, for movement within channel **127**. Top end **137** (FIG. 12) includes a cylindrical aperture **139** for receiving cam pin **135** (FIG. 11) of cam head **85**. Top end **137** rotates about cam pin **135**. FIGS. 10 and 11 illustrate the shaft **56** being rotatably connected to cam pin **135**.

[0050] As shown in FIG. 12, shaft **56** includes an additional cylindrical aperture **141** which passes through top end **137**. Aperture **141** is located orthogonal to aperture **139**, and aperture **141** is positioned to intersect with aperture **139**, as shown in FIG. 12. Aperture **141** is sized to receive a lock pin **143**, as shown in FIG. 11.

[0051] Referring to FIG. 13, the lever **58** is assembled, by initially positioning the top end **137** of shaft **56** into channel **127** of cam head **85**. Cam pin **135** is next passed through apertures **133** (FIG. 3) in each of side members **123**, **125** of the cam head and into aperture **139** (FIG. 12) of the shaft head. Finally, lock pin **143** (FIG. 13) is passed through aperture **141** (FIG. 12) located in the top end of shaft **56** and is passed through an aperture **145** in cam pin **135**.

[0052] Thus, shaft **56** is held in position by both the aperture **139** in the shaft as well as the lock pin **143**. Lock pin **143** secures the shaft **56** to the cam pin **135**, and thus increases the wear surface between shaft **56** and the cam head **85**.

[0053] Referring to FIG. 14, hole passages **70**, **72** of the two clamps **52**, **54** are shaped in an offset manner. Particularly, the wall thickness at **151** above each of hole passages **70**, **72** and the wall thickness at **153** below each of hole passages **70**, **72** is set at .070 inches. Whereas, the wall thickness at **155** between the hole and the distal end of the respective clamps is .100 inches. This, slight offset of the hole position serves to prevent the collapse of the hole due to continued use of the clamp without rods located in the hole passages **70**, **72**, e.g., without a retractor arm **44** (FIG. 1) or extension arms **19**, **20** or cross bar **18** or post **17**, within hole passages **70**, **72**.

[0054] As shown in FIG. 15, a second multi-directional joint clamp **211** includes a first clamp **213**

and a second clamp **215**. Joint clamp **211** includes a central shaft **217** as well as a turn-handle **219** for bringing first clamp **213** and second clamp **215** into a clamping position or into an unclamping position. Central shaft **217** includes a threaded portion **221** that mates with a second threaded portion (not shown) formed in the turn-handle **219**. As the turn-handle **219** is rotated, shaft **217** is drawn upwardly into the turn-handle **219** moving the two clamps **213**, **215** together in order to perform the clamping function.

[0055] First clamp **213** includes a passage **221** which is intended to accommodate, for example, post **17** of rail clamp **12** (FIG. 2). Similarly, second clamp **215** includes a passage **223**. Passage **223** is intended to accommodate, for example, cross bar **18** (FIG. 2). Passages **221**, **223** may be shaped in an offset manner similar to passages **70**, **72** as described above in relation to FIG. **14**.

[0056] Referring now to FIG. **16**, a third multi-directional joint clamp **321** is shown. The third multi-directional joint clamp **321** may be implemented in a manner similar to the multi-directional joint clamp **21** of FIGS. **3-14**. However, the joint clamp **321** of FIG. **16** uses a different type of stop for the central shaft. In particular, the joint clamp **321** may include a central shaft **356** with an integral or fixed head **390** at its distal end instead of a nut. The head **390** may engage a shim **392**, which may adjust a distance **D2** between a top surface **391** of the washer **357** and a top surface **394** of the integral head **390**. A thicker shim **392** increases the distance **D2** and thereby increases the pressure load placed on the clamps **352**, **354** by the cam lever **358** when in a clamping position since the length of the shaft **356** remains unchanged. Conversely, a thinner shim **392** decreases the distance **D2** and thereby decreases the pressure load placed on the clamps **352**, **354**. Thus, a desired pressure load may be obtained by selecting a shim **392** having an appropriate thickness. The multi-directional joint clamp **21** of FIGS. **3-14** may also adjust a pressure load applied by its cam lever **58**. However, the multi-directional joint clamp **21** may achieve such result by tightening or loosening the nut **113** instead of or in addition to using a shim **392**.

[0057] While particular embodiments of the invention have been shown, it will be understood that the invention is not limited thereto since modifications may be made by those skilled in the art, particularly in light of the foregoing teaching. It is, therefore, the appended claims which define the true spirit and scope of the invention.

Claims

1-20. (canceled)

21. A surgical retraction system, comprising: one or more frame members; a plurality of retractor blades, each retractor blade comprising a retractor arm and a blade portion at a distal end of the retractor arm; and a plurality of joint clamps that couple a proximal portion of each respective retractor arm to the one or more frame members, wherein each joint clamp of the plurality of joint clamps includes: an upper clamp comprising an upper clamp top side, an upper clamp bottom side, and an upper clamp passage configured to circumscribe a received proximal portion of a respective retractor arm, wherein the upper clamp top side comprises an upper clamp aperture with a tapered inner surface; a lower clamp comprising a lower clamp top side, a lower clamp bottom side, and a lower clamp passage configured to circumscribe a received frame member of the one or more frame members; a washer comprising a washer top surface and a washer lower surface that mates with the tapered inner surface of the upper clamp aperture; a shaft comprising a shaft upper portion and a shaft lower portion, wherein the shaft extends through the washer, the upper clamp aperture, the upper clamp, and the lower clamp; a handle coupled to the shaft upper portion; and a lower device secured to the shaft lower portion, wherein the lower device permits adjusting a magnitude of a clamping force applied between the handle and the lower device when the handle is rotated to a clamping position.

22. The surgical retraction system of claim 21, wherein: the clamping force constricts the upper clamp passage around its received proximal portion of the respective retractor arm; and the

clamping force constricts the lower clamp passage around its received frame member of the one or more frame members.

23. The surgical retraction system of claim 21, wherein: the shaft lower portion comprises threads; and the lower device comprises threads that engage the threads of the shaft lower portion.

24. The surgical retraction system of claim 23, wherein: rotation of the lower device in a first direction reduces a distance between the lower device and the handle; and rotation of the lower device in a second direction opposite the first direction increases the distance between the lower device and the handle.

25. The surgical retraction system of claim 21, wherein the lower device is welded to the lower shaft portion to retain the clamping force at a desired magnitude.

26. The surgical retraction system of claim 21, wherein: the handle is coupled to a shaft upper portion via a cam head; and an outer surface of the cam head applies the clamping force when the handle is rotated to the clamping position.

27. The surgical retraction system of claim 26, wherein a flat surface of the outer surface of the cam head engages the washer when the handle is rotated to the clamping position.

28. The surgical retraction system of claim 21, wherein the upper clamp and the lower clamp are each rotatable about an axis of the shaft.

29. The surgical retraction system of claim 21, wherein: the lower clamp bottom side comprises a lower clamp aperture with a tapered inner surface; and an upper surface of the lower device mates with the tapered inner surface of the lower clamp aperture.

30. The surgical retraction system of claim 21, wherein the upper clamp bottom side comprises teeth that engage teeth of the lower clamp top side.

31. The surgical retraction system of claim 21, wherein: the tapered inner surface of the upper clamp aperture comprises a conical surface; and the washer comprises a conical surface that mates with the conical surface of the upper clamp aperture.

32. A surgical retraction system, comprising: one or more frame members; a plurality of retractor blades, each retractor blade comprising a retractor arm and a blade portion at a distal end of the retractor arm; and a plurality of joint clamps that couple a proximal portion of each respective retractor arm to the one or more frame members, wherein each joint clamp of the plurality of joint clamps includes: an upper clamp comprising an upper clamp top side, an upper clamp bottom side, and an upper clamp passage, wherein the upper clamp top side comprises an upper clamp aperture with a tapered inner surface; a lower clamp comprising a lower clamp top side, a lower clamp bottom side, and a lower clamp passage; a washer comprising a washer top surface and a washer lower surface that mates with the tapered inner surface of the upper clamp aperture; a shaft comprising a shaft upper portion and a shaft lower portion, wherein the shaft extends through the washer, the upper clamp aperture, the upper clamp, and the lower clamp; a handle coupled to the shaft upper portion; and a lower device secured to the shaft lower portion, wherein the lower device permits adjusting a magnitude of a clamping force applied between the handle and the lower device when the handle is rotated to a clamping position, wherein the upper clamp passage is configured to circumscribe a received proximal portion of a respective retractor arm or a received frame member of the one or more frame members; and wherein the lower clamp passage is configured to circumscribe a received proximal portion of a respective retractor arm or a received frame member of the one or more frame members.

33. The surgical retraction system of claim 32, wherein: the clamping force constricts the upper clamp passage around its received proximal portion of the respective retractor arm or its received frame member of the one or more frame members; and the clamping force constricts the lower clamp passage around its received proximal portion of the respective retractor arm or its received frame member of the one or more frame members.

34. The surgical retraction system of claim 32, wherein: the shaft lower portion comprises threads; and the lower device comprises threads that engage the threads of the shaft lower portion.

35. The surgical retraction system of claim 32, wherein: rotation of the lower device in a first direction reduces a distance between the lower device and the handle; and rotation of the lower device in a second direction opposite the first direction increases the distance between the lower device and the handle.

36. The surgical retraction system of claim 32, wherein the lower device is welded to the lower shaft portion to retain the clamping force at a desired magnitude.

37. The surgical retraction system of claim 32, wherein: the handle is coupled to a shaft upper portion via a cam head; an outer surface of the cam head applies the clamping force when the handle is rotated to the clamping position; and a flat surface of the outer surface of the cam head engages the washer when the handle is rotated to the clamping position.

38. The surgical retraction system of claim 32, wherein: the lower clamp bottom side comprises a lower clamp aperture with a tapered inner surface; and an upper surface of the lower device mates with the tapered inner surface of the lower clamp aperture.

39. The surgical retraction system of claim 32, wherein: the upper clamp and the lower clamp are each rotatable about an axis of the shaft; and the upper clamp bottom side comprises teeth that engage teeth of the lower clamp top side.

40. The surgical retraction system of claim 32, wherein: the tapered inner surface of the upper clamp aperture comprises a conical surface; and the washer comprises a conical surface that mates with the conical surface of the upper clamp aperture.
